

Clement LEE

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Academic statistician with expertise in statistical programming and visualisation in R, and modelling of extremes and networks with computational Bayesian statistics.

WORK EXPERIENCE

Lecturer in Statistics 2022-
School of Mathematics, Statistics and Physics, Newcastle University
Research and teaching. Past administrative role: seminar organiser.

Lecturer in Statistics 2019-2022
Department of Mathematics and Statistics, Lancaster University
Research and teaching. Award: ***Tower of Teaching*** for outstanding teaching in the department.

Research Associate (Statistical Modeller) 2016-2019
School of Mathematics and Statistics, Newcastle University
Undertaken statistical research using data generated by mobile and web technologies developed by the Open Lab, with a focus on social network and citation network analysis.

Research Associate 2015
Department of Mathematics and Statistics, Lancaster University
Derived the optimal scaling of one Markov chain Monte Carlo (MCMC) sampler, and implemented efficient MCMC algorithms for stochastic epidemic models based on such optimal scaling.

Actuarial Intern 2010
BOC Group Life Assurance Company Ltd., Hong Kong
Performed statistical analysis to assist pricing and valuation, and helped with checking the rates of new insurance products.

Assistant in Actuarial Science 2009
Department of Statistics and Actuarial Science, the University of Hong Kong
Developed an Excel software of insurance pricing for teaching and consulting.

EDUCATION

PhD in Statistics 2011-2015
Lancaster University
Thesis title: Statistical Models for Financial Time Series and the House Price Index.

MSc in Statistics 2010-2011
Lancaster University
Passed with Distinction. Awards: ***Chancellor's Medal***, and ***Royal Statistical Society prize*** for outstanding performance.

BSc in Actuarial Science

2006-2009

The University of Hong Kong

1st Class Honours. Awards: *HKU Worldwide Student Exchange Scholarship*, and *HKU Foundation Entrance Scholarship*.

PUBLICATIONS

A Bayesian Nonparametric Stochastic Block Model for Directed Acyclic Graphs

2026

C. Lee and M. Battiston

Journal of Computational and Graphical Statistics **35** (2), 619–630

How to Utilise the Limited Supply of Vaccines for Mpx Control in Thailand among High-Risk GBMSM

2026

S. Jirapanakorn, B. Hunsin, A. Endo, C. Lee, S. Jin, B. S. L. Dickens and T. Theparod

Infectious Disease Modelling **11** (3), 1009–1021

Evidencing Preferential Attachment in Dependency Network Evolution

2025

C. Lee

arXiv preprint arXiv:2509.12135

Learning Growth Mechanisms of Tail Realistic Preferential Attachment Models from Network Degree Distributions

2025

T. Boughen, C. Lee and V. Palacios Ramirez

arXiv preprint arXiv:2506.18726

Software Version Control Using Git: A Primer from UKRN

2025

C. Lee et al.

OSF Preprint

Conditional Extremes with Graphical Models

2024

A. Farrell, E. F. Eastoe and C. Lee

arXiv preprint arXiv:2411.17013

Degree Distributions in Networks: Beyond the Power Law

2024

C. Lee, E. Eastoe and A. Farrell

Statistica Neerlandica **78** (4), 702–718

The Hierarchical Stochastic Block Model for Multiple Networks

2024

M. Battiston and C. Lee

Submitted

Estimating the Asphaltene Critical Nanoaggregation Concentration Region Using Ultrasonic Measurements and Bayesian Inference

2021

A. Svalova, D. Walshaw, C. Lee, V. Demyanov, N. G. Parker, M. J. Povey and G. D. Abbott

Scientific Reports **11**, 6698

A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets

2020

C. Lee and D. J. Wilkinson

Journal of the American Statistical Association **115** (529), 1–15

Performance and Sensitivities of Home Detection on Mobile Phone Data	2020
M. Vanhoof, <u>C. Lee</u> and Z. Smoreda Big Data Meets Survey Science: A Collection of Innovative Methods	
A Review of Stochastic Block Models and Extensions for Graph Clustering	2019
<u>C. Lee</u> and D. J. Wilkinson Applied Network Science 4 (1), 122	
Validity and Reliability of an Online Self-Report 24-h Dietary Recall Method (Intake24): A Doubly Labelled Water Study and Repeated-Measures Analysis	2019
E. Foster, <u>C. Lee</u> , F. Imamura, S. E. Hollidge et al. Journal of Nutritional Science, 8	
Weaving the Topics of CHI: Using Citation Network Analysis to Explore Emerging Trends	2019
<u>C. Lee</u> , A. Garbett, J. Wang, B. Hu and D. Jackson Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems, 6	
A Network Epidemic Model for Online Commissioning Data	2018
<u>C. Lee</u> , A. Garbett and D. J. Wilkinson Statistics and Computing 28 (4), 891-904	
Optimal Scaling of the Independence Sampler: Theory and Practice	2018
<u>C. Lee</u> and P. Neal Bernoulli 24 (3), 1636-1652	
A Social Network Analysis of Articles on Social Network Analysis	2018
<u>C. Lee</u> and D. J. Wilkinson arXiv preprint arXiv:1810.09781	
ThinkActive: Designing for Pseudonymous Activity Tracking in the Classroom	2018
A. Garbett, D. Chatting, G. Wilkinson, <u>C. Lee</u> and A. Kharrufa Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, 7	
LearningCircle.io: Lessons Learned from Organising Courses with and without a Dedicated Platform	2018
H. Celina, <u>C. Lee</u> , P. Olivier and A. Kharrufa EdMedia+ Innovate Learning, 76-81	

FUNDING SUCCESSES

Knowledge Exchange Placement	2026
<i>Funder: DICE Network+</i> Project: Predicting item desirability with machine learning to boost digital platform exchanges	
Knowledge Exchange Catalyst	2025
<i>Funder: International Centre for Mathematical Sciences</i> Project: Turbo-charging the gift economy through geospatial optimisation	

PHD SUPERVISION

Clive Drowley

2024-

Geospatial Centre for Doctoral Training, Newcastle University

First supervisor; Topic on geospatial analysis of network flows

Thomas Boughen

2023-

School of Mathematics, Statistics and Physics, Newcastle University

First supervisor; Topic on intersections of extreme value theory, networks & Bayesian nonparametrics

Aiden Farrell

2021-2025

Department of Mathematics and Statistics, Lancaster University

Second supervisor; Topic on the extremal behaviour of the nodes when a network grows

Maarten Vanhoof

2017-2018

School of Computing, Newcastle University

Second supervisor; Thesis title: Geographical Veracity of Indicators from Mobile Phone Data

TEACHING

Undergraduate and Postgraduate Teaching

2023-

School of Mathematics, Statistics and Physics, Newcastle University

Taught modules for BSc and MSc students in Mathematics, Statistics & Data Science, including *Statistical Foundations of Data Science*, *Statistical Learning Methodology*, and *Modern Bayesian Inference*. Developed and delivered new module on *Data Visualisation*.

Postgraduate Teaching

2019-22

Department of Mathematics and Statistics, Lancaster University

Taught modules for MSc students in Statistics and in Data Science, including *R programming*, *Statistical Inference*, *Generalised Linear Models*, and *Extreme Value Theory*.

Graduate Teaching Assistant

2011-2015

Department of Mathematics and Statistics, Lancaster University

Marked assignments, held workshop sessions, and provided help and feedback to students. Modules included *Bayesian Inference*, *Likelihood Inference*, *Time Series Analysis*, and *Stochastic Processes*.

PROFESSIONAL ACTIVITIES

Statisticians for Society Scheme

2024-

Royal Statistical Society

Member of review panel for *pro bono* consultancy projects carried out by fellow RSS statisticians for charity organisations.

Local Lead on Open Research

2024-

UK Reproducibility Network (UKRN) & Newcastle University

Led grassroots activities on open research supported by UKRN, at Newcastle University, including the university's inaugural Open Research Conference, training workshops on computational reproducibility, and a journal club on open research.

INVITED TALKS

Community-Powered Circular Economy: The Role of Data <i>Exeter Centre for Circular Economy</i>	2026-05-14
Evidencing Preferential Attachment in Dependency Network Evolution <i>Statistics Seminar, Department of Mathematics, University of York</i>	2026-03-19
Breaking Barriers: Enhancing ECR Engagement in Open Science <i>ReproducibiliTea Journal Club, University of York</i>	2025-03-28
Lessons from My (S)crappy Reproducibility Journey <i>Royal Statistical Society 2024 Annual Conference</i>	2024-09-04
Citation Network Analysis of a Computer Science Conference <i>The Bartlett Centre for Advanced Spatial Analysis, UCL</i>	2020-02-28
A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets <i>Department of Statistics and Actuarial Science, University of Hong Kong</i>	2019-01-02

CONFERENCES

Sunbelt 2025 Conference <i>Paris, France</i> Talk: <i>Evidencing Preferential Attachment in Dependency Network Evolution</i>	2025
Sunbelt 2024 Conference <i>Edinburgh, UK</i> Talk: <i>From the Power Law to Extreme Value Mixture Distributions</i>	2024
useR! 2021 Conference <i>Online</i> Virtual talk: <i>The Evolution of the Dependencies of CRAN Packages</i>	2021
Bayesian Inference in Stochastic Processes Workshop <i>Online</i> Virtual talk: <i>A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets</i>	2021
Royal Statistical Society 2019 Annual Conference <i>Belfast, United Kingdom</i> Talk: <i>Clustering the Citation Network of a Computer Science Conference</i>	2019
COSTNET Early Career Training Event <i>Munich, Germany</i> <i>Best Poster Award</i> for <i>A Social Network Analysis of Articles on Social Network Analysis</i>	2019
23rd International Conference on Computational Statistics <i>Iasi, Romania</i> Talk: <i>A Social Network Analysis of Articles on Social Network Analysis</i>	2018
COSTNET17 Conference on Statistical Network Science	2017

Palma de Mallorca, Spain

Poster: *A Network Epidemic Model for Online Community Commissioning Data*

8th Conference on Extreme Value Analysis

2013

Fudan University, Shanghai

Talk: *Modelling Non-stationary Extremes and Threshold Uncertainty*

SKILLS

<i>Programming</i>	R, C++ and Python (elementary)
<i>Text editing</i>	EMACS, L ^A T _E X, and (R)Markdown
<i>Miscellaneous</i>	Git and Bash
<i>Languages</i>	English, Cantonese Chinese (native), Mandarin Chinese (fluent)